proportional to the distance it has been displaced. We call the force a *restoring force* because it pulls the mass back toward the equilibrium position.

Another example is a marble rolling back and forth at the bottom of a bowl, with no energy being lost to friction. What characterizes these systems is a potential energy function that looks like a parabola:

$$V(x) = \frac{k}{2}x^2. \qquad (10.1)$$

The constant *k* is called the *spring constant*. If were call that the force on an object is minus the gradient of *V*, we find that the force on the object is

$$F = -kx. \qquad (10.2)$$

The negative sign tells us that the force acts opposite to the displacement and pulls the mass back toward the origin.

Why are harmonic oscillators so prevalent in physics? Because almost any smooth function looks like a parabola close to a minimum of the function. Indeed, many kinds of systems are characterized by an energy function that can be approximated by a quadratic function of some variable representing a displacement from equilibrium. When disturbed, these systems will all oscillate about the equilibrium point. Here are some other examples:

- An atom situated in a crystal lattice. If the atom is displaced slightly from its equilibrium position, it gets pushed back with an approximately linear restoring force. This motion is three-dimensional and really consists of three independent oscillations.

- The electric current in a circuit of low resistance often oscillates with a characteristic frequency. The mathematics of circuits is identical to the mathematics of masses attached to springs.